

SUDEEP K S

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Available: January – August 2027

EDUCATION

Northeastern University, Boston, MA

Candidate for a Master of Science in Applied Artificial Intelligence

Expected May 2028

GPA: 3.89/4.0

Related courses: Machine Learning, Deep Learning, Natural Language Processing, Applied Generative AI

IIIT Bangalore, Bengaluru, India

Advanced Certificate, Data Science (Deep Learning)

Sep 2025

CGPA: 3.89/4.0

BMS Institute of Technology & Management, Bengaluru, India

Bachelor of Engineering

2022

CGPA: 7.2/10

TECHNICAL KNOWLEDGE

Core Skills: Data Visualization (Tableau, Matplotlib, Seaborn), Statistical Evaluation & Benchmarking, Deep Learning, NLP, Computer Vision, LLMs, RAG

Languages: Python, SQL

Frameworks & Libraries: Pandas, GeoPandas, PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, OpenCV, FastAPI, Streamlit

Cloud & DevOps: Azure, Docker, MLflow, Git

WORK EXPERIENCE

Capgemini, Bengaluru, India

Aug 2023 – Dec 2025

Senior Software Engineer — RPA & Automation | Client: Humana, Inc. (health insurance & wellness services)

- Automated rule-based processes for claims processing, letter generation, and HRA (Health Risk Assessment) workflows using Automation Anywhere 360, reducing production defects by 30% through systematic edge-case coverage.
- Partnered with business analysts to gather requirements and author end-to-end RPA solution designs (SDD/ADD) using Visio and Lucidchart across 10+ automated pipelines.
- Directed RPA delivery across 5 agile sprint teams using Azure DevOps CI/CD, incorporating unit/integration testing and UAT to reduce deployment errors by 25% and ensure functional and compliance requirements were met.
- Wrote SQL queries against Humana's internal data warehouse to retrieve and validate member eligibility data (contract ID, PBP ID, member status) across Medicaid, Medicare, and commercial plans, then confirmed pass/fail execution status via process/transaction ID queries across 10+ automated pipelines.
- Monitored production bot performance across 3 platforms, resolved issues through root-cause analysis, maintained 98%+ ServiceNow ticket resolution within SLA, and mentored junior engineers on RPA best practices.

Rinex Technologies, India

Sep 2022 – Jul 2023

Front-End Developer

- Built 8+ responsive forms and UI pages for Rinex's e-learning platform using HTML5, CSS, JavaScript, Bootstrap, and PHP, improving page load performance by 40% through code optimization and lazy loading.
- Collaborated closely with a UI/UX designer to translate design mockups into functional, pixel-accurate front-end layouts.
- Implemented multi-page navigation flows and PHP-based form handling to support core platform interactions.
- Worked within a cross-functional development team to plan, build, and ship features end-to-end on the platform.

PROJECTS

HealthGuard AI — Multimodal Clinical Intelligence Platform

LangGraph / LLMs / Computer Vision / RAG • <https://github.com/kssudeep/healthguard-ai>

- Architected a production-grade 5-agent LangGraph system (Vision, NLP, RAG, Critic, Synthesizer) using a supervisor-reflection pattern with circuit breaker logic, generating end-to-end clinical reports in ~12 seconds.
- Fine-tuned DenseNet-121 on the NIH ChestX-ray-14 dataset (14 pathologies) with Grad-CAM explainability for interpretable diagnostic predictions.
- Implemented BioBERT-based NER for clinical entity extraction with negation detection using SpaCy, improving structured extraction from unstructured clinical text.
- Built a hybrid FAISS + BM25 retrieval pipeline with cross-encoder reranking over PubMed and clinical guideline corpora, improving retrieval relevance by 31% over a single-method baseline.
- Deployed to Azure Container Apps using Docker, with MLflow experiment tracking (26 metrics/run), Prometheus monitoring, and Redis caching across 5 containerized services.

OmegaTK RAG Coding Assistant — Commissioned by OpenEye / Cadence Molecular Sciences

LangChain / FAISS / RAG / Gemini • <https://github.com/kssudeep/omegatkg-rag>

- Built a RAG-based coding assistant for OpenEye's OmegaTK Python toolkit using FAISS + Gemini 2.5 Flash, generating executable molecular conformer code from natural language queries.
- Engineered a history-aware, two-step RAG chain (query condensation + MMR retrieval, k=8) over 8 live documentation pages, achieving 15/15 on an automated test suite.
- Delivered a Streamlit chat interface with conversation history, inline source citations, and index rebuild controls for end users.
- Instrumented a full observability layer with stage-level tracing, p50/p95 latency tracking, and cost-per-request logging via structured JSONL logs, enabling production-grade monitoring of RAG chain performance.

Optimising NYC Taxi Operations — Exploratory Data Analysis

Python / Pandas / GeoPandas / Seaborn / Statistical Analysis

- Performed EDA on a year of NYC yellow taxi trip records (stratified hourly 5% sampling, random_state=42), cleaning missing values, merging duplicate columns, and removing outliers via the IQR method.
- Merged trip records with NYC Taxi Zone shapefile data using GeoPandas for zone- and borough-level spatial analysis of pickup/dropoff patterns.
- Analyzed hourly, daily, monthly, and quarterly demand and revenue trends, identifying peak pickup windows (4–8 PM) and top zones such as JFK Airport, Midtown, and Upper East Side.
- Computed correlation and fare-per-mile analyses across distance, duration, and vendor to surface pricing and routing insights.